(Please write your Enrollment Number) **Enrollment No._____**

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech (IT/AIML/CSE/CSE-AI)>, Semester:<03>
(December, 2024)

Subject Code: BIT 203	**Subject: Software Engineering**
Time :3 Hours	**Maximum Marks :60**

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1	**Attempt all parts**	**20**	
	a) 'Think Big, Act Small'. Enumerate two reasons for software projects going wrong?	**2.5**	**CO1**
	b) List the Umbrella Activities that occur throughout the Software Process.	**2.5**	**CO1**
	c) List some metrics (property and measure) for specifying non-functional requirements	**2.5**	CO2
	d) Developing a software application is intrinsically complex. Justify your answer	**2.5**	**CO2**
	e) Discuss design concept "Separation of Concern".	**2.5**	**CO3**
	f) Contrast between coupling and cohesion	**2.5**	**CO3**
	g) What is estimation method in project planning? Briefly explain the key ingredients of an estimation method?	**2.5**	**CO4**
	h) Discuss the concept of 'Size Equivalent' in Detailed COCOMO model.	**2.5**	**CO4**
UNIT I			
Q2	Software agile methods are designed to produce useful software quickly. Describe some common characteristics shared by Agile methods.	**(10)**	**CO1**
Q3	Compare and contrast between Prescriptive Process Models and Evolutionary Process Models.	**(10)**	**CO1**
UNIT II			
Q4	A student Registration System for a University is a digital platform designed to streamline and manage the student registration for subject being offered in a semester. At the beginning of the semester, student is required to register for the compulsory course offered by the Department. In addition, student has to select fixed number of optional courses out of the elective course offered by the Department. Every elective course has a fixed number of seats A course can be compulsory courses or the elective courses offered in the discipline of study. As a system engineer, you are required to present the student registration process a) Using context/zero level DFD b) First Level DFD	**(3,7)**	**CO2**
Q5	In literature, a number of Requirements elicitation techniques have been proposed. Why one elicitation technique cannot work in all situations?	**(10)**	**CO2**
UNIT III			

Q6	Consider the following Program:
	
	`def check_triangle(a, b, c):`
	`  if a == b:`
	`  if a == c:`
	`  if b == c:`
	`  return "Equilateral"`
	`  else:`
		(4,3, 3)	**CO3**

	return "Isosceles" **else**: **return** "Isosceles" **else**: **if** b != c: **if** a == c: **return** "Isosceles" **else**: **return** "Scalene" **else**: **return** "Isosceles" (i) Draw CFG (ii) Find cyclomatic complexity (iii) Identify Independent paths and write test cases for each independent path.		
Q7	Prepare the structured chart for the following module: 1. BEGIN 2. Get count 3. names[] = loadList (count) 4. 5. i = 0 6. WHILE i < length of names 7. Get newName 8. duplicate = checkForDuplicate (names, newName) 9. IF duplicate == False THEN 10. addToList (names, newName) 11. ENDIF 12. i = i + 1 13. ENDWHILE 14. END 15. 16. BEGIN checkForDuplicate (names, newName) 17. namesSorted = sortNames (names) 18. inList = findName (namesSorted, newName) 19. Return inList 20. END mySubFour	**(10)**	
	UNIT IV		
Q8	A software manager is in charge of the development of a safety-critical software system, which is designed to control a radiotherapy machine to treat patients suffering from cancer. This system is embedded in the machine and must run on a special-purpose processor with a fixed amount of memory (256 Mbytes). The machine communicates with a patient database system to obtain the details of the patient and, after treatment, automatically records the radiation dose delivered and other treatment details in the database. The COCOMO method is used to estimate the effort required to develop this system and an estimate of 26 person-months is computed. All cost driver multipliers were set to 1 when making this estimate. Explain why this estimate should be adjusted to take project, personnel, product and organizational factors into account. Suggest four factors that might have significant effects on the initial COCOMO estimate. Justify why you have included each factor?	**(10)**	**CO4**
Q9	Explain FPA model used for software estimation with suitable example.	**(10)**	**CO4**